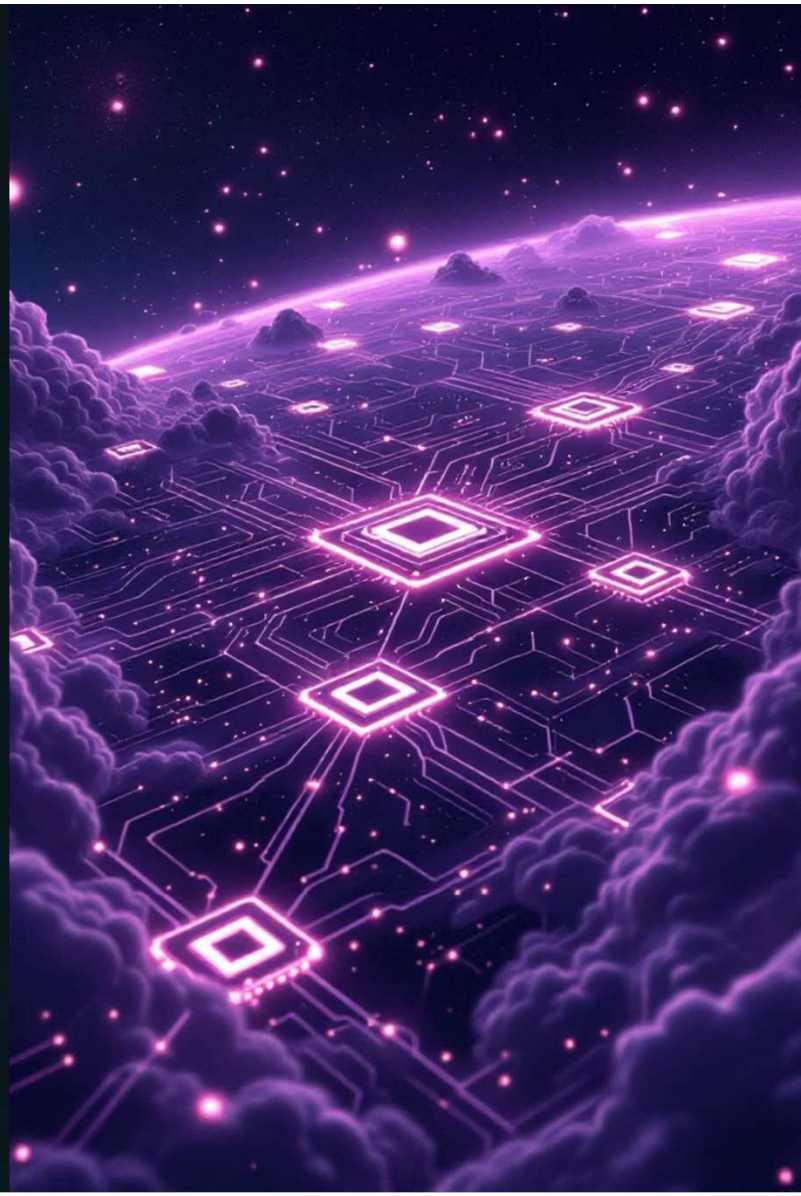
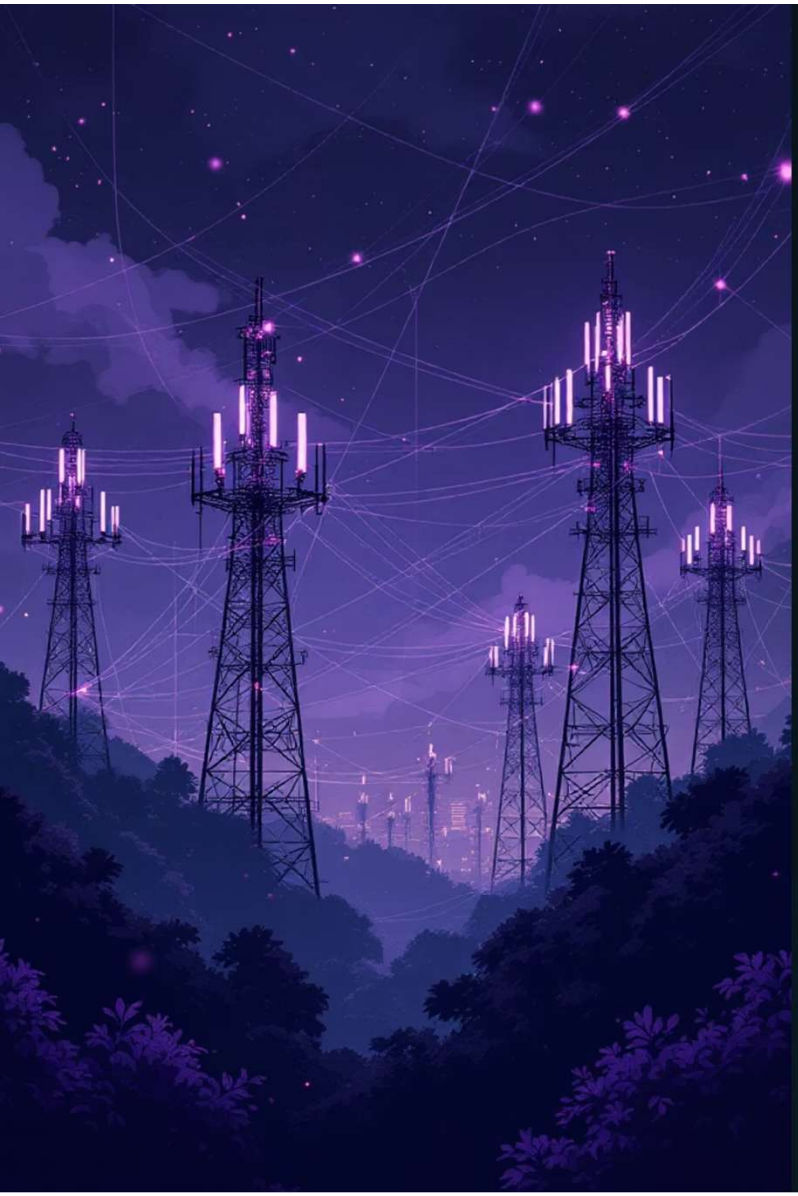
Security Threats and AI Defenses in 6G

A literature review based on 20 research papers

INFO 3180 – Wireless Networks · Jodhveer Atwal · Khushi Malik · Mary Klassen · Arda Yazici · Victor Mai





Why This Topic Matters

6G is AI-Native

AI will drive resource allocation, traffic optimization, anomaly detection, and autonomous decision-making — not just the network.

A Dual Attack Surface

Attackers can now target both the **network** and the **AI models** themselves.

Takeaway: In 6G, security is no longer only a network problem — it is also a machine learning problem.

Background

Massive Scale + Complexity

Ultra-low latency, massive IoT connectivity, intelligent automation, and integration of terrestrial, aerial, and satellite systems.

O-RAN

Multi-vendor, software-based systems

More flexible, but larger attack surface

AI is in the Core of the Network

Used for routing, optimization, and automation

Decisions are made **without human input**

Trust is No Longer Assumed

Zero Trust → everything must be verified

XAI → decisions must be explainable

Xu 2022; Kaur & Gupta 2025

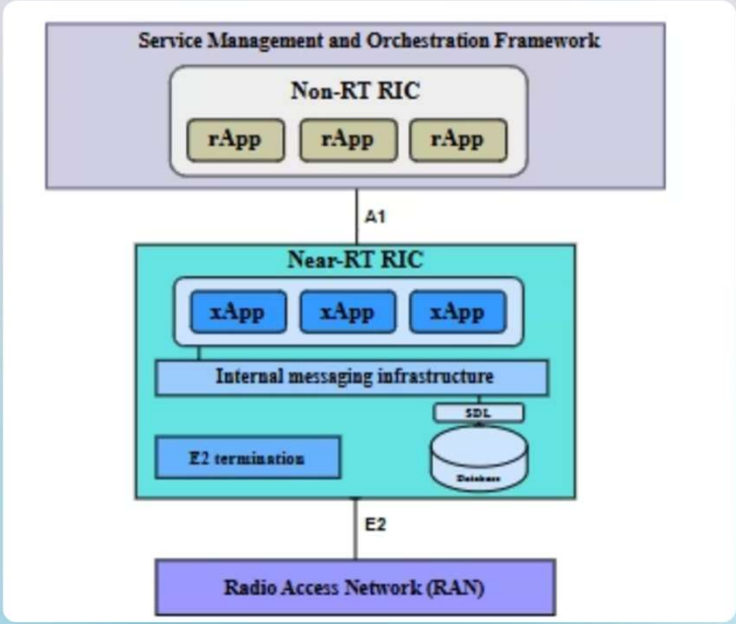


Figure 1: Simplified O-RAN architecture

Background

Network Layers

Terrestrial + satellite + relay nodes

Integrated Network with Universal Coverage

Increased Attack Surface

More links → more entry points

More nodes → harder to secure

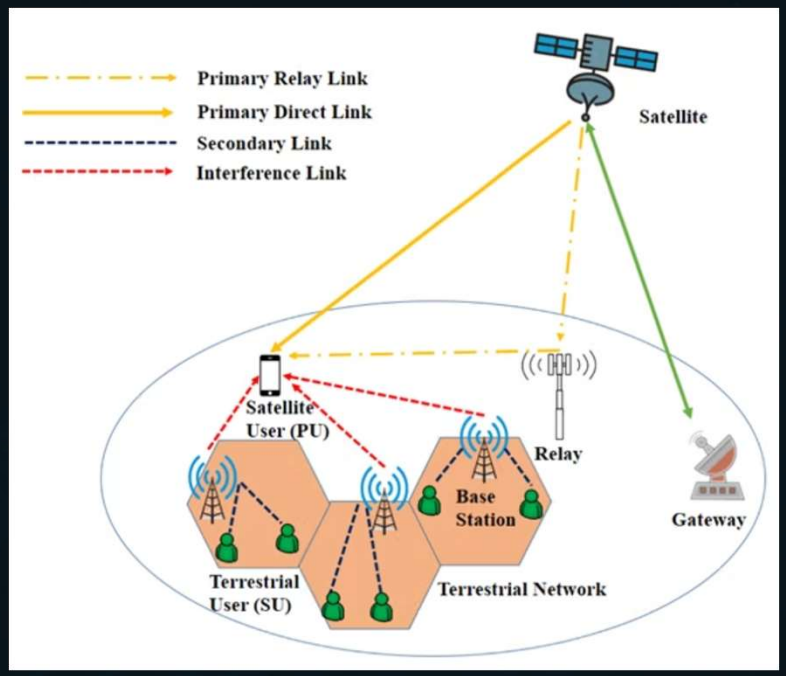


Figure 2: basic visualization of how a 6g network setup

Bountakas 2023; Xu 2022; Chiejina 2024

Main Security Threats in AI-Driven 6G

Evasion Attacks

Manipulate input data at inference time, causing the AI to make wrong decisions.

Poisoning Attacks

Corrupt training data so the model learns incorrect behavior from the start.

Privacy Stealing

Infer private information from model updates shared during federated training.

O-RAN & Edge Risks

xApps and rApps can be manipulated; weak or malicious edge participants degrade model accuracy.

Takeaway: The AI pipeline itself becomes part of the attack surface.

Shafahi 2018; Catak 2021; Chiejina 2024; Xu 2022

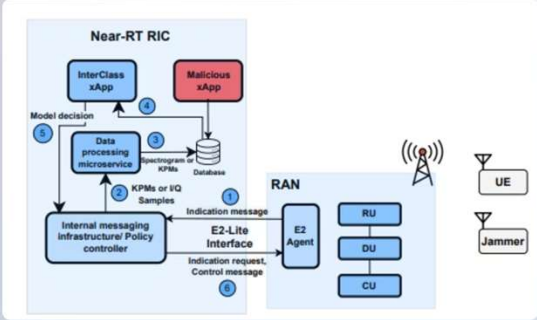


Figure 3: Overview of Interference Classification xApp (with Malicious xApp).

Examples from the Literature

Clean-Label Poisoning
Malicious samples appear normal but still corrupt model behavior.

mmWave Beam Attacks
Small signal perturbations misdirect wireless beams, degrading connectivity.

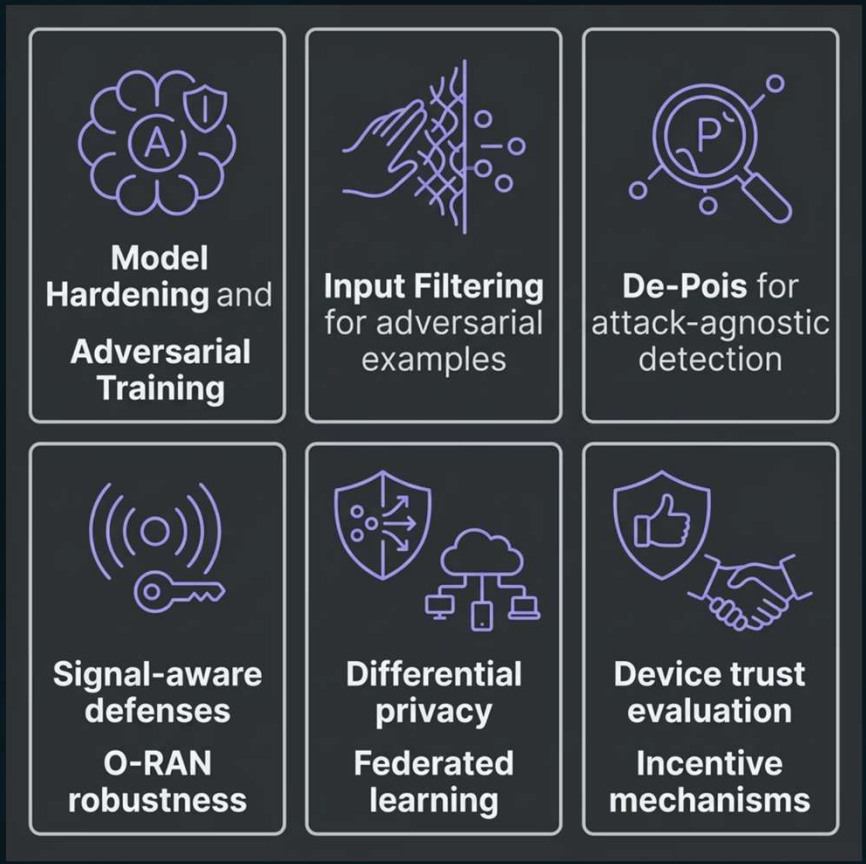
O-RAN Adversarial Attacks
Manipulated spectrograms and KPIs reduce AI model accuracy in radio networks.

FL Risks in 6G
Cheating attacks, low-quality local training, and privacy stealing undermine federated learning.

Takeaway: These are not theoretical — they directly affect performance, reliability, and trust.

Bountakas 2023; Chen 2021; Xu 2022

Current Defense Landscape



Promising but Fragmented

Defenses span adversarial training, poisoning detection, differential privacy, and trust evaluation — but are often tested only in narrow, isolated settings.

Takeaway: The tools exist. The challenge is integrating them into a cohesive, deployable system.

Bountakas 2023; Xu 2022; Chiejina 2024

The Key Gap in the Literature

Siloed Solutions

Most papers address only **one** dimension: robustness, poisoning detection, privacy, trust, or explainability — rarely all together.

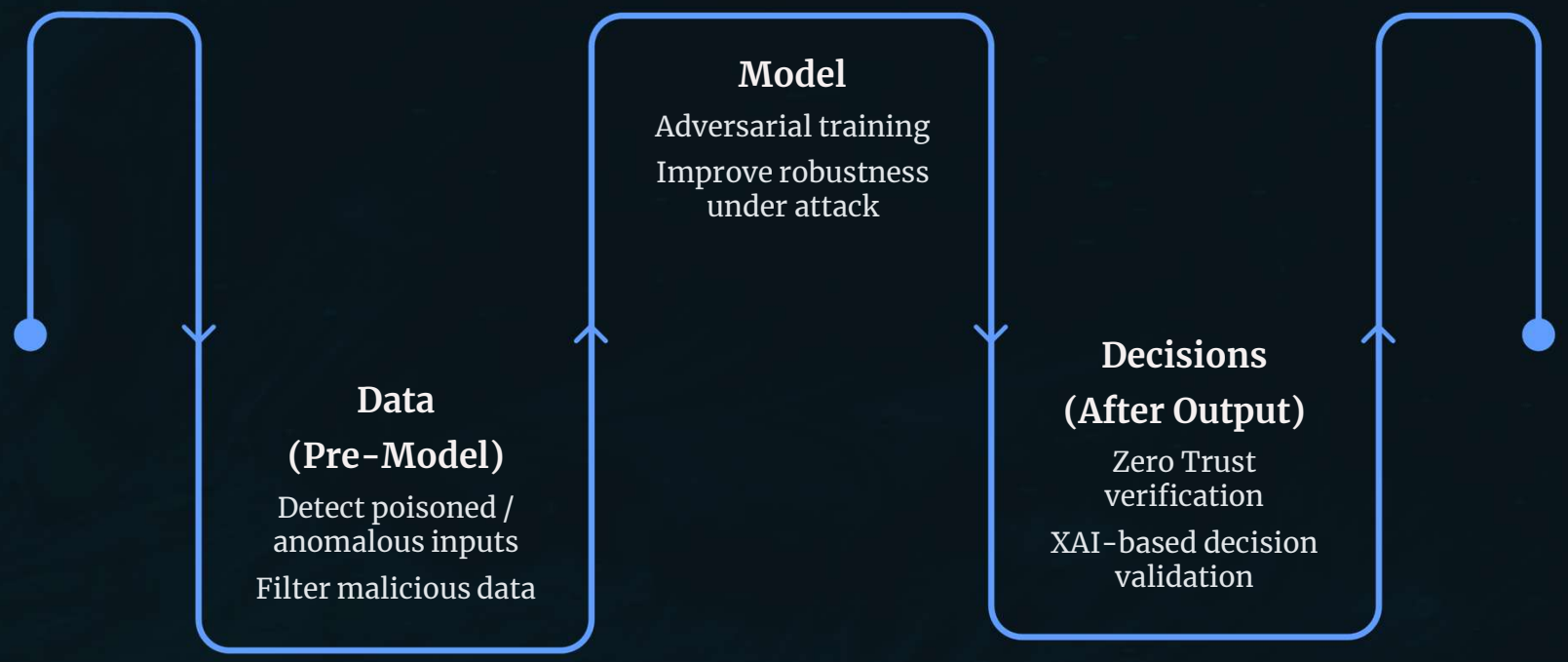
Current Limitations

- Not always scalable
- Not always explainable
- Not fully tested in realistic 6G environments
- May add latency or computational overhead

Main Gap: 6G needs a defense framework that is **integrated, practical, and deployment-ready**.

Team synthesis based on Chen 2021; Catak 2021; Aramide 2024; Ali 2025

Proposed: Three Phase Defense Framework



Security is distributed across all stages, reducing single points of failure.

Proposed: Three Phase Defense Framework

Data Phase

- Anomaly detection / filtering
- Mimic model (De-Pois)
- Goal: detect poisoning before training

Model Phase

- Adversarial training
- Exposure to perturbed inputs
- Goal: improve robustness

Decision Phase

- Zero Trust verification
- XAI (SHAP, LIME)
- Goal: validate and interpret outputs

Conclusion

AI is Both Enabler and Target

6G depends on AI for intelligent networking — making it a critical asset and a prime attack vector.

Threats Are Real

Evasion, poisoning, privacy leakage, and O-RAN manipulation pose direct risks to 6G reliability.

Future Direction

Layered, explainable, trust-aware, real-time AI defense frameworks are the path forward.

Final message: A secure 6G network must defend not only its infrastructure, but also the intelligence that controls it.

Key References

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Defense Strategies for Adversarial Machine Learning: A Survey

02

Security and Privacy in Artificial Intelligence-Enabled 6G

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System-level Analysis of Adversarial Attacks and Defenses on Intelligence in O-RAN based Cellular Networks

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De-Pois: An Attack-Agnostic Defense Against Data Poisoning Attacks

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Poison Frogs! Targeted Clean-Label Poisoning Attacks on Neural Networks

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A Survey on Zero Trust Architecture for 6G

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Cyber-Security in the 6G Era: Emerging Threats, Multi-Layered Defense, and Future Directions